SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – CASE STUDY ANALYSIS

Course Code	**CSA10 / CSA1021**	**Course Title**	**Software Engineering**
CO Assessed	CO1	Assessment	Assessment Tool 1 – Case Study Analysis
Weightage	20%	Total Marks	25
CO1 Statement	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name	P.PramodKumar	Reg. No.	1925260245

Date: _____ Duration: 60 minutes

Code of Conduct

I ___P.Pramod Kumar (1925260245)Name / Reg. No.) certify that this submission is my original work
and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____P.Pramod Kumar_____

Annexure A – Case Study Analysis

Instructions to Students

Each student will be allotted one problem statement. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how modularity, scalability, maintainability, reliability, usability, and security are incorporated.

5. Justify the Chosen Methodology/Framework

- Select a suitable SDLC model or framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why it is suitable for the proposed system.
- Explain how it supports successful development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe expected benefits.
- Explain how the solution addresses identified challenges.
- Discuss risks, implementation challenges, and limitations.
- Suggest future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, roles and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses and all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution unclear/inappropriate; requirements largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Innovative, feasible solution applying appropriate principles such as modularity, scalability, maintainability and security.	Suitable solution with good application of software engineering principles.	Basic solution with limited application of principles.	Methodology inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/framework and provides strong technical justification.	Suitable methodology with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations or future enhancements missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/Framework Justification (10%)	/5

TOTAL: _____ /25

Industry Context

Job markets are moving away from traditional four-year degrees toward continuous skills development and micro-credentials. However, job applicants often struggle to compile, share, and verify diverse certifications earned from online courses, bootcamps, and workplace training programs. Employers face slow, expensive verification steps when hiring. The modern educational ecosystem needs a secure, lifelong learning registry where individuals can verify their skills instantly.

Problem Statement

Design and develop a decentralized Lifelong Learning Ledger that allows educational institutions, employers, and online platforms to issue unalterable micro-credentials. The platform must manage a secure registry of skill badges, course completions, and professional certifications. It must allow students to aggregate these milestones into a digital portfolio and share them with recruiters for instant verification without needing a central school registrar.

Project Aim

Objectives

- Build a decentralized platform to issue, claim, and organize digital skill credentials.
- Implement cryptographic validation to detect document tampering, forgery, or invalid issuer signatures.
- Create secure sharing controls so learners decide which credentials are visible to each recruiter.
- Integrate with human-resources hiring tools to automate skill verification during job applications.
- Support flexible micro-credential updates as technologies, professional standards, and job roles evolve.

Expected Outcomes

- Fast, low-cost verification of professional skills for hiring managers and recruiters.
- Improved career mobility through a verified portfolio of lifelong learning achievements.
- Reduced administrative effort for schools, training providers, and workplace academies.
- A clearer, data-driven understanding of workforce skills and education trends, using aggregated and privacy-protected information.

To propose a practical, privacy-conscious credential platform that makes learning achievements portable and easier to verify across institutions, training providers, employers, and online learning services.

1. Understand the Problem Statement

1.1 Problem in Our Own Words

People now gain skills through universities, short online courses, bootcamps, professional training and workplace learning. These achievements are often stored in separate accounts or paper certificates. When a person applies for a job, the employer may need to contact each provider and wait for confirmation. Records can be lost, difficult to compare, or altered. The proposed Lifelong Learning Ledger provides a trusted way to issue, collect, share and verify digital credentials without depending on one central registrar for every check.

1.2 Objectives

- Allow approved universities, training providers and employers to issue digitally signed micro-credentials.
- Maintain tamper-evident records of credential references, issuer identity, issue time and status.
- Give learners a digital portfolio to collect course completions, skill badges and certifications.
- Let learners share selected credentials with recruiters through a link or QR code.
- Enable quick verification of issuer signature, credential integrity, expiry and revocation status.
- Protect personal information by sharing only the information needed for a particular purpose.

1.3 Scope

The proposed scope includes issuer onboarding, credential creation and signing, learner portfolios, consent-based sharing, recruiter verification, revocation/status checks, access management and audit logs. A distributed ledger can store a credential hash, issuer identifier, timestamp and status pointer, while the credential itself can be held securely by the learner or a protected storage service. This reduces unnecessary public exposure of personal data.

The system does not itself guarantee that a course provider teaches well or that every credential represents the same level of skill. Issuer accreditation, assessment quality and credential definitions remain the responsibility of trusted organizations. The ledger also does not replace recruitment decisions; it provides evidence that can support them.

1.4 Success Criteria

- Credential verification succeeds for valid credentials and detects altered, revoked or incorrectly signed credentials.
- Learners can share a selected credential without exposing their entire portfolio.
- Verification remains available under expected usage and has a measured response time.
- Only authorized issuers can create credentials, and issuer keys are protected.
- The system records status changes and administrative actions for accountability.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders and Responsibilities

Stakeholder	Role / responsibility
Learners / students	Receive credentials, maintain a portfolio, choose what to share and request corrections when information is

	wrong.
Universities and colleges	Confirm completion of approved programs and issue credentials under institutional policy.
Online course and bootcamp providers	Issue badges or certificates for verified course completion and skills assessments.
Employers and recruiters	Request permission to view credentials and check authenticity, issuer and current status.
Professional certification bodies	Issue credentials that meet defined professional standards and manage renewal or expiry.
Platform administrators	Manage onboarding, service health, permissions, backups and incident response; they should not be able to forge issuer signatures.
Ledger / infrastructure operators	Maintain distributed infrastructure and availability according to the network governance model.
Auditors and regulators	Review compliance, access controls, retention and evidence of proper operation.

2.2 Major Challenges

- Trust and governance: the network must determine which issuers are authorized and how they are removed or suspended.
- Privacy: credentials may contain names, grades, dates or other personal details that should not be permanently exposed on a public ledger.
- Interoperability: institutions may use different data formats, naming conventions and identity systems.
- Key security: a stolen issuer signing key could be used to create fraudulent credentials until it is revoked.
- Revocation and correction: expired, withdrawn or incorrectly issued credentials need a clear status process.
- Adoption and cost: providers need simple tools and clear incentives to join the system.
- Accessibility: learners and recruiters need a usable service across devices and assistive technologies.

2.3 Risk Mitigation

Use a governed issuer registry, strong authentication, role-based permissions, key rotation, secure backups and independent security reviews. Keep personal credential data off-chain where practical; place only a hash or minimal status reference on the ledger. Use standard credential schemas and provide correction, expiry and revocation procedures. Pilot the system with a small group of issuers before expanding.

3. Analyze the Current Approach

3.1 Existing Process

Many organizations issue PDF certificates, paper documents, badges hosted on separate platforms or records stored in student-information systems. Recruiters may inspect a certificate manually, email the issuing institution, use a provider-specific verification page or request official transcripts. These methods can work, but the process differs by issuer and often requires repeated manual effort.

3.2 Strengths

- Established institutions already maintain official student and completion records.
- Signed PDFs and provider verification pages can help confirm some certificates.
- Central student-information systems can support corrections and controlled access.
- Existing digital badge services may provide shareable links and structured metadata.

3.3 Limitations and Gaps

- Records are fragmented across institutions, course platforms and employers.
- Manual email checks can be slow and expensive, particularly for international or older records.
- A certificate image or PDF can be edited or copied unless its authenticity is checked.
- Provider-specific portals create multiple accounts and inconsistent verification experiences.
- A central database may become a single point of failure or control.
- Credential status changes may not be visible to a recruiter who only has an old downloaded copy.

3.4 Comparison Table: Current and Proposed Approach

Aspect	Typical current approach	Proposed decentralised ledger
Storage	Separate portals, PDFs and institutional databases	Learner-held digital portfolio; ledger stores tamper-evident references/status
Verification	Manual email, portal lookup or visual inspection	Automated signature, integrity and status check
Speed	May take hours or days when manual checks are required	Designed for near-instant digital verification; actual speed must be tested
Tamper detection	Depends on the issuer's portal or manual confirmation	Credential signature and ledger hash can reveal unauthorized changes
Privacy	Documents may be sent in full by email	Learner shares selected credentials and minimum necessary data
Revocation	May require contacting issuer or checking a separate portal	Status/revocation endpoint can be checked during verification
Governance	Each provider controls its own process	Defined issuer admission, ledger governance and dispute procedures
Limitations	Fragmented records and repeated effort	Requires adoption, interoperable standards, secure key management and governance

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Proposed System

The proposed platform uses a permissioned or carefully governed distributed ledger for tamper-evident credential references. Each approved issuer creates a structured digital credential and signs it using a protected private key. The learner receives the credential in a portfolio. When a recruiter requests proof, the learner shares a credential or a limited presentation link. The verifier checks the issuer's signature, the credential's integrity and its current status. The system returns a clear result such as valid, revoked, expired or unable to verify.

A decentralised ledger is not the only way to build credential verification, so the design should justify its use against a well-governed shared database. The ledger is useful where multiple independent issuers need a shared tamper-evident record and no single organization should control the whole registry. Sensitive personal details should not be written directly to an immutable public chain.

4.2 Main Functional Modules

Issuer registration and governance: Onboards institutions and certification providers; verifies their authority and maintains issuer status.

Credential issuance: Creates a structured credential with achievement, issuer, date, learner reference and optional expiry; signs it digitally.

Ledger registry: Records credential hash, issuer identifier, timestamp and status reference to make later tampering detectable.

Learner portfolio: Stores and organizes credentials from multiple providers and supports search, grouping and export.

Consent and sharing: Creates a time-limited link or QR code and allows learners to select which credentials to disclose.

Verification service: Checks signature, hash, issuer status, expiry and revocation state, then returns an understandable result.

Revocation and correction: Allows authorized issuers to revoke an invalid credential and issue a corrected replacement.

Administration, audit and reporting: Manages roles, security events, availability metrics and authorized audit reports.

4.3 Functional Requirements

- The system shall authenticate issuers and verify that they are authorized to issue a credential type.
- It shall create digitally signed credentials using a documented schema.
- It shall store a tamper-evident credential reference and status without exposing unnecessary personal information.
- It shall allow learners to collect credentials from multiple approved issuers.
- It shall support consent-based sharing with recruiters through a secure link or QR code.
- It shall verify signatures, issuer status, integrity, expiry and revocation state.
- It shall support revocation, correction and re-issuance by authorized issuers.
- It shall provide role-based administration, audit logs and monitoring.

4.4 Non-Functional Requirements

- Security: encryption in transit and at rest, strong authentication, least-privilege access and protected signing keys.

- Privacy: data minimization, learner consent, appropriate retention and no unnecessary personal information on-chain.
- Performance: verification should be fast enough for recruitment workflows; establish and test a measurable service-level target.
- Availability: redundant services, monitoring, backups and recovery procedures.
- Scalability: support increasing numbers of learners, issuers and verification requests.
- Interoperability: use documented schemas and APIs so different providers can exchange credentials.
- Usability and accessibility: provide clear status messages, mobile-friendly screens and accessible navigation.
- Maintainability: modular code, automated tests, documentation and controlled releases.

4.5 Software Engineering Principles

Modularity separates issuance, portfolio, ledger, verification and administration so that one component can be updated without rebuilding the whole platform. Scalability is supported by stateless APIs, caching of public issuer keys and independently scaled verification services. Maintainability comes from documented interfaces, versioned credential schemas and automated tests. Reliability requires backups, health checks and recovery procedures. Security is built into authentication, key storage, access control and audit logging. Usability is improved through simple learner sharing controls and recruiter results that explain what was verified.

4.6 Real Workflow Diagram

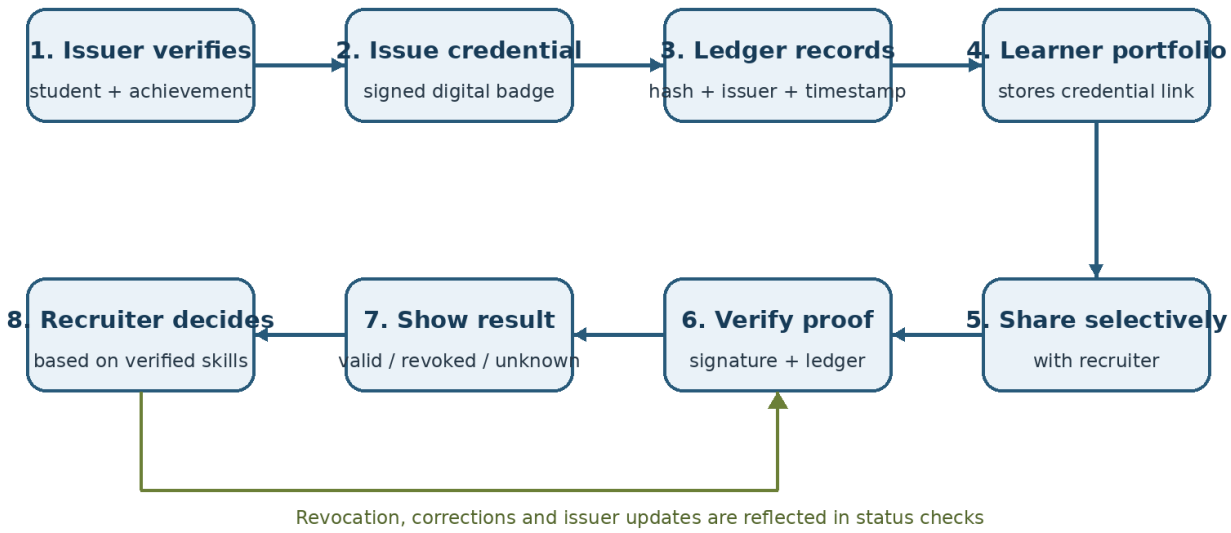


Figure 1. Proposed credential issuance, learner sharing and recruiter verification workflow.

4.7 Example Credential Verification Matrix

Check	What the system checks	Possible result
Issuer identity	Is the issuer registered and active?	Trusted issuer / unknown or suspended issuer
Digital signature	Was the credential signed by the	Signature valid / invalid

	issuer's valid key?	
Integrity	Does the credential match its recorded hash?	Unchanged / altered
Status	Has the issuer revoked or replaced it?	Active / revoked / replaced
Validity period	Is it expired, where expiry applies?	Current / expired
Learner consent	Is the recruiter permitted to view the shared data?	Authorized / access denied
Overall outcome	Combine all checks; do not rely on one signal alone	Valid / invalid / revoked / expired / unable to verify

4.8 Feasibility Study

4.8.1 Technical Feasibility

The system is technically feasible as a staged web platform built from established components: a browser-based learner portal, issuer dashboard, verification API, credential database, cryptographic signing service, and a permissioned ledger or append-only registry. Standard web technologies can support authentication, APIs, audit logging, and QR-based verification. A small pilot can begin with a limited number of issuers and a modest credential volume before introducing distributed consensus or integrations at larger scale. Technical feasibility depends on choosing mature libraries, documenting the credential schema, protecting issuer keys, and testing interoperability early.

4.8.2 Operational Feasibility

The platform fits existing education and recruitment workflows if it does not force providers to replace their student-information systems. Institutions can upload or issue credentials through a dashboard or API; learners can claim them in a portfolio; recruiters can scan a QR code or open a consent-based link. Adoption will require short training sessions, clear help material, a support process for incorrect records, and a published policy for issuer onboarding and revocation. The interface should be mobile-friendly and accessible to users with different levels of digital confidence.

4.8.3 Economic Feasibility

The main costs are software development, cloud hosting, security testing, key management, integration work, staff training, and ongoing support. A pilot can control cost by using open standards and open-source components where appropriate, limiting the first release to essential features, and avoiding storage of large files or personal records on a ledger. Benefits may include fewer manual verification emails, shorter hiring delays, reduced certificate re-issue work, and less duplicate administration. These benefits should be measured during the pilot rather than assumed in advance.

4.8.4 Legal, Privacy, and Ethical Feasibility

The project is feasible only if data protection and education-record obligations are addressed before deployment. Personal information should remain off the ledger wherever possible; the ledger should store only a hash, issuer reference, timestamp, and status pointer. Learners need understandable consent controls, a way to correct inaccurate data through the issuer, and a clear explanation of what a verifier can see. A privacy impact review, retention policy, breach-response plan, and legal review for each deployment region should be completed before a live launch.

4.8.5 Schedule and Resource Feasibility

A practical student-project prototype can be planned over approximately 10 to 12 weeks: requirements and stakeholder interviews in weeks 1–2; interface and data-model design in weeks 3–4; issuer and learner modules in weeks 5–7; verification, status checking, and security controls in weeks 8–9; pilot testing and corrections in weeks 10–11; and documentation and demonstration in week 12. The schedule assumes a small cross-functional team and may need adjustment if external institutional integrations are unavailable.

4.8.6 Feasibility Summary

Overall, a controlled pilot is feasible, while a nationwide or cross-border network would require more extensive governance, legal review, interoperability testing, and operational funding. The recommended decision is to proceed with a limited prototype, validate user demand and verification

accuracy, and expand only when the pilot meets agreed privacy, security, usability, and performance criteria.

4.9 Architecture, Data Handling, and Governance

The architecture should separate the learner interface, issuer interface, verification service, identity and access service, credential store, and ledger adapter. This separation keeps the system maintainable and permits the ledger technology to be changed without rewriting the user-facing applications. Each credential should have a unique identifier, issuer identity, achievement description, issue date, optional expiry, learner-controlled presentation, digital signature, and status reference. The verifier checks the signature and issuer trust status, confirms that the credential has not been revoked, and returns a clear result with the checks performed.

Governance is as important as technology. A consortium policy should define who may issue credentials, the evidence required for issuer approval, how signing keys are rotated, how disputes are handled, and who can suspend an issuer. A credential should not be treated as trustworthy merely because a record exists on a ledger; the verifier must also confirm the issuer's authority and the quality or meaning of the underlying assessment.

4.10 Pilot Plan and Acceptance Criteria

The first pilot should involve a small set of participating providers, a limited group of learners, and a recruitment team willing to test verification. The pilot should use sample or consented records, document every test case, and avoid placing sensitive personal data on a public network. Acceptance criteria should include successful verification of valid credentials, rejection of altered or incorrectly signed credentials, visible warnings for revoked credentials, appropriate access restrictions, and successful sharing of selected credentials without revealing the learner's entire portfolio.

The pilot should also record baseline and post-pilot measures: median and 95th-percentile verification time, number of manual follow-up requests, successful verification rate, false acceptance and false rejection rates, user task completion, support tickets, and cost per verification. Targets should be agreed with stakeholders before testing; results should be reported honestly, including failed tests and limitations.

4.11 Technology Choice: Distributed Ledger versus Central Registry

The project should avoid selecting a distributed ledger simply because it is fashionable. The choice depends on governance and trust requirements. A central registry is usually simpler and cheaper when one trusted organization controls the service. A permissioned ledger can be justified when multiple independent issuers need shared write permissions, auditability, and a common status record. In either case, the system still needs identity checks, digital signatures, privacy controls, backups, and a dispute process.

Decision factor	Central registry	Permissioned distributed ledger
Governance	One accountable operator	Shared rules among approved organizations
Implementation cost	Usually lower	Usually higher due to node operation and governance
Tamper evidence	Depends on access controls and audit logs	Shared history can make unauthorized changes easier to

		detect
Privacy	Straightforward centralized access management, but concentrated data risk	Requires careful design; personal data should still remain off-ledger
Availability	Can depend on one operator unless replicated	Can tolerate some node failures if configured correctly
Best fit	Single trusted authority or small consortium	Independent organizations needing a common audit trail

5. Justify the Chosen Methodology / Framework

5.1 Selected Methodology: Agile with Scrum

Agile with Scrum is suitable because the platform involves several user groups, privacy requirements, identity integrations and evolving credential standards. The team can deliver a small working feature set, collect feedback from learners and issuers, and improve it in short sprints. Security and interoperability risks can be tested early instead of being left until the end.

5.2 How Scrum Supports the Project

- Product backlog: prioritize issuer onboarding, credential issuance, learner portfolio, sharing and verification.
- Sprint planning: agree a realistic sprint goal and acceptance criteria.
- Daily Scrum: coordinate engineering, security, design and testing work.
- Sprint review: demonstrate working features to representative learners, issuers and recruiters.
- Retrospective: identify process problems and improve the next sprint.
- Continuous integration: run automated tests and security checks whenever code changes.

5.3 Illustrative Sprint Plan

Sprint	Focus	Deliverable
1	Requirements, stakeholder interviews, threat model and architecture	Approved backlog, user journeys and architecture decision
2	Issuer registration, identity and role controls	Issuer onboarding prototype
3	Credential schema, signing and issuance API	First signed test credential
4	Learner portfolio and consent-based sharing	Portfolio with shareable credential presentation
5	Ledger reference, verification and status checking	End-to-end verification prototype
6	Revocation, accessibility, security and performance testing	Tested release candidate and known-issues report
7	Pilot, feedback, documentation and deployment preparation	Pilot report and prioritized improvements

5.4 Testing and Maintenance

Testing should include unit tests for credential creation and status logic, integration tests between issuer and verifier services, signature and tampering tests, access-control tests, privacy reviews, usability tests and load tests. Before deployment, the team should test key loss, issuer suspension, service outage and revocation scenarios. After release, monitor availability, verification failures, response time and user feedback. Updates to schemas or signing keys should be versioned and rolled out with a rollback plan.

6. Discuss Expected Outcomes and Limitations

6.1 Expected Outcomes

- Less manual work for recruiters who need to confirm learning achievements.
- Faster verification of signed credentials from participating issuers.
- One learner portfolio for achievements earned across different platforms.
- Better detection of altered certificates through signature and integrity checks.
- Clearer credential status through expiry and revocation checks.
- More learner control over which credentials are shared with each recruiter.
- Improved auditability through issuer records and traceable status changes.

6.2 Risks, Limitations and Controls

Risk / limitation	Why it matters	Control or response
Untrusted issuer	A technically valid credential can still represent poor-quality training.	Vet issuers, publish issuer criteria and show issuer identity clearly.
Lost or stolen signing key	Attackers may issue false credentials under a provider's identity.	Use hardware-backed key protection where feasible, rotation and rapid revocation.
Privacy leakage	Immutable public data can expose personal details permanently.	Keep personal data off-chain; store hashes/status references and obtain consent.
Incorrect source data	The ledger cannot prove that the original assessment was fair or accurate.	Require issuer quality controls, correction processes and independent audits.
Interoperability problems	Different providers may use incompatible formats.	Adopt shared schemas, documented APIs and versioning.
Network or service outage	Verification may be delayed or unavailable.	Use redundant services, monitoring, backups and a documented fallback.
Adoption and cost	Small providers may lack resources to integrate.	Offer simple APIs, a lightweight issuer portal and phased onboarding.

6.3 Evaluation Measures

The pilot should measure median and 95th-percentile verification time, percentage of valid credentials verified successfully, detection of modified credentials, revocation update delay, service availability, learner completion of sharing tasks, recruiter satisfaction and issuer onboarding time. Baselines and target values should be agreed before testing. No measured improvement is claimed in this proposal because no implementation or pilot dataset was supplied.

6.4 Future Enhancements

- Support recognized verifiable credential formats and interoperable digital wallets.
- Add selective disclosure so a learner can prove a qualification without revealing unrelated details.
- Enable cross-border issuer trust lists and credential translations.
- Provide accessible mobile applications and offline-friendly credential presentation.
- Introduce analytics for skills gaps while using aggregated or privacy-protected data.
- Develop formal governance for issuer disputes, network upgrades and long-term data availability.

6.5 Implementation Risk Matrix

Risk	Likelihood / impact	Mitigation	Owner
Issuer signing key compromised	Medium / High	Hardware-backed key storage where practical, multi-person recovery, rotation, monitoring, and rapid issuer suspension.	Security lead
Incorrect credential issued	Medium / High	Issuer review, confirmation workflow, auditable correction and replacement process.	Credential issuer
Personal data disclosed unnecessarily	Medium / High	Data minimization, off-ledger storage, consent-based sharing, privacy review, and access logging.	Privacy officer
Low participation by providers	Medium / Medium	Pilot with willing institutions, provide simple onboarding, document cost and benefits, support standard APIs.	Product owner
Service outage or unavailable status endpoint	Medium / Medium	Backups, monitoring, recovery drills, replicated services, and clear temporary-unavailable status.	Operations team

6.6 Overall Evaluation

The project should be judged on trustworthy verification, learner control, interoperability, accessibility, and operational sustainability—not on the presence of blockchain alone. A conventional signed-credential service with a shared status registry may be sufficient for some deployments; a distributed ledger is most useful when several independent organizations need a common, tamper-evident record and no single party should control the registry. The final architecture should follow the results of the feasibility study and a transparent comparison of cost, governance, privacy, and reliability.

Conclusion

The Decentralised Lifelong Learning Ledger and Microcredential Verification Matrix addresses the fragmentation and slow verification of learning achievements across universities, online courses, bootcamps and professional training. By combining digitally signed credentials, a learner-controlled portfolio, tamper-evident ledger references and automated status checks, the proposed system can make credential verification more convenient and transparent. A distributed ledger is most valuable when independent issuers need a shared record and clear governance; it should not be treated as proof that the original course or assessment was high quality. Privacy, issuer trust, key security, revocation and interoperability must be designed from the beginning. Agile with Scrum supports incremental delivery and feedback from learners, providers and recruiters. A controlled pilot with measurable performance, security and usability criteria is recommended before wider deployment.

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THANK YOU